MIL-R-36675A
1 OCTOBER 1986
SUPERSEDING
MIL-R-36675
15 May 1970

MILITARY SPECIFICATION

RETRACTOR, MASTOID

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of self-retaining mastoid retractor of the Weitlaner design, suitable for the retraction of the scalp in surgery.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 S. 20th Street, Philadelphia, PA 19101, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6515

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SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Box, Fiberboard.
PPP-B-676	- Boxes, Set-up.
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-L-10547	- Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-794	- Parts and Equipment, Procedures for Packaging of.

2.1.2 Drawing. The following drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

DRAWING

Defense Personnel Support Center (DPSC)

24231 - Retractor, Mastoid, Weitlaner, 3 Versus 4 Prongs

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.

E 92 - Vickers Hardness of Metallic Materials, Tests for.

E 140 - Hardness Conversion Tables for Metals (Relationship Between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness and Knoop Hardness).

F 899 - Stainless Steel Billet, Bar and Wire for Surgical Instruments.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The mastoid retractor shall be in accordance with the applicable requirements of GG-I-526, and as specified herein.

3.2 Material. The mastoid retractor shall be fabricated from Class 4 martensitic stainless steel in accordance with ASTM F899.

3.3 Hardness. The Rockwell hardness of the mastoid retractor shall be C40 to C45.

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3.4 Style, design and dimensions. Shall be a self-retaining mastoid retractor of the Weitlaner design used to retract the scalp during surgery. The retractor shall have 3 vs 4 blunt prongs. Shall be capable of successive retraction at each ratchet catch without unlocking. The operation of the retractor shall be easy, without drag or binding. The style, design, and dimensions of the mastoid retractor shall conform to Defense Personnel Support Center (DPSC) Drawing No. 24231.

3.5 Finish. Shall be mirror, satin or matte finish. The matte finish shall exhibit a uniform nondirectional surface texture equivalent to that produced by lightly blasting the prepared surface with five abrasives under pressure.

3.5.1 Passivation. The retractor shall be properly passivated.

3.6 Operation test.

3.6.1 Operation of ratchet catch mechanism. When tested as specified in 4.4.2.1, the ratchet catch shall engage and disengage the ratchet teeth smoothly. There shall be no binding or hesitation.

3.6.2 Engagement of ratchet teeth by ratchet catch mechanism. When tested as specified in 4.4.2.2, the retractor shall open and not unlock at each place checked by closing. There shall be no hesitation or binding.

3.6.3 Operation of retractor without locking. Action shall be smooth and easy when tested as specified in 4.4.2.3. Shall not bind in any position.

3.7 Identification markings. In addition to the requirements of GG-I-526, instruments of foreign manufacture shall also be marked "stainless."

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, or does it commit the Government to acceptance of defective material.

4.1.2 Inspection. Inspection shall be in accordance with GG-I-526.

4.1.3 Certificate of quality. Certificate of quality supplied by the manufacturer of the stainless steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the item identification, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in table I. Unit of product for examination purposes shall be one retractor.

TABLE I. Sampling for examination

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test purposes shall be one retractor. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

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TABLE II. Sampling for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Composition Analysis	3.2 and G-G-526	GG-I-526	**	*
Hardness test	3.3 & GG-I-526	GG-I-526	S-2	1.0
Dimensions	3.4	---	S-2	1.5
Copper sulfate test	GG-I-526	GG-I-526	S-2	1.5
Boil test	GG-I-526	GG-I-526	S-2	1.5
Operation test	3.6	4.4.2	S-2	1.5

* Acceptance number shall be zero for all sample sizes.

** Sufficient amount from each lot of each metal used in the construction of the mastoid retractor shall be selected for composition analysis.

4.3 Examination.

4.3.1 Examination of retractor. The mastoid retractor shall be examined to determine compliance with all the requirements. Examination shall be conducted in accordance with MIL-STD-105 and the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in Table III.

TABLE III. Classification of defects

Categories	Defects
Critical	None defined.
Major	
101	Style and design not as specified.
102	Instrument not constructed as specified.
103	Instrument not free from sharp edges, rough areas, burrs, cracks, grind marks, or overlaps.
104	Instrument binds or unlocks while being opened.
Minor	
201	Finish not as specified.
202	Instrument not permanently and legibly marked with either the name or trademark of the contractor.

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4.4 Tests. Tested shall be conducted to determine compliance with specification requirements.

4.4.1 Hardness test. Hardness shall be determined in accordance with ASTM Standards E 18 or E 92. Conversion of Vickers hardness values to Rockwell values shall be in accordance with ASTM Standard E 140.

4.4.2 Operation test.

4.4.2.1 Operation of ratchet catch mechanism. Without operating the retractor, fully disengage the ratchet catch mechanism from the ratchet teeth so the spring is displaced. Repeat 5 times.

4.4.2.2 Engagement of ratchet teeth by ratchet catch mechanism. Starting with the prongs fully closed, open the retractor completely. Stop at 4 or 5 places to close the retractor. Note degree of engagement between ratchet teeth and ratchet catch mechanism.

4.4.2.3 Operation of retractor without locking. Disengage catch mechanism from the ratchet teeth. Starting with the retractor arms completely open, close the retractor arms until the prongs are together (closed). Return to open position and repeat three times. Note ease of action.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Retractors shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Each retractor shall be packaged in accordance with MIL-P-116, Method IC-2.

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5.1.1.4 Intermediate package. Six retractors shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for retractors shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Eighteen intermediate packages (108 retractors) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Eighteen intermediate packages (108 retractors) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636 class domestic. Closure shall be as specified in the box specification.

5.2.3 Level C. The level C packing for retractors shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in a exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Retractors packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.2.5.2 Level C. Retractors packed as specified in 5.2.3 shall be unitized as specified in 5.2.5.

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5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Intended use. The mastoid retractor is intended for the retraction of the scalp in surgery for mastoid.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) National Stock Number.

6.3 This specification covers the following National Stock Number (NSN):

NSN	Item Identification
6515-00-926-9193	RETRACTOR, MASTOID, Weitlaner, 3 vs 4 Blunt Prongs

This specification does not include all types which are commercially available.

6.4 Subject term (key word) listing.

Mastoid retractor
Retractor

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MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-3486

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-36675A	RETRACTOR, MASTOID

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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